Sparsh Mittal

Curriculum Vitae

Employment

Associate professor at ECE department, IIT Roorkee from 08/2024 onwards. Also, a joint faculty at Mehta Family School of Data Science (DS) and Artificial Intelligence (AI) at IIT Roorkee. IEEE Senior member

Assistant professor at ECE department, IIT Roorkee from 12/2019 to 08/2024.

Assistant professor at CSE department, IIT Hyderabad from 09/2016 to 12/2019.

Postdoctoral Research Associate at Oak Ridge National Lab (USA) from 09/2013 to 09/2016

Current research areas: Artificial intelligence (AI) for computer-vision, applications of AI/ML/DL, accelerators for AI, computer architecture, VLSI, approximate computing.

Education

Ph.D., Computer Engineering , 2008-2013, Iowa State University (ISU), USA.
Thesis topic: *Dynamic Cache Reconfiguration Based Techniques for Improving Cache Energy Efficiency*

B.Tech., Electronics and Communications Engineering 2004-2008
Indian Institute of Technology (IIT) Roorkee, Uttarakhand, India. GPA: 9.58/10.0

Honors and Awards

I received IIT Roorkee's "Excellence in Teaching Award" 2025 in the UG category (for classes with 80+ students).

I was shortlisted in the list of top-5 faculty members for IITR's excellence in teaching award 2024 in two categories: UG course with more than 80 students and UG course with more than 30 students (taught by a young faculty member).

I received "2nd Ramesh Mittal Award for Excellence in Research" from the Roorkee branch of Bharat Vikas Parishad. Bharat Vikas Parishad has more than 2 lakh members in India and its Roorkee branch has won the best NGO award in Roorkee.

"Best Paper Award" at VLSID (India) 2026, APCCAS (South Korea) 2025 and COMSNETS India Internet Governance Workshop (India) 2024.

"Best Industry-Related Paper" Award at ICPR 2024 conference (India).

"Best paper award honorable mention" at AIMLSystems Conference (Bengaluru) 2023

"Best student paper award" at VLSI Design Conference (Kolkata) 2024.

"Best Poster Award" at AIMLSystems (Bengaluru) 2025 to two of my posters.

Best student paper finalist in SC (Supercomputing) 2014, world's biggest HPC conference with 10,000+ attendees.

Finalist in Gyan Setu competition held in Gyan Bharatam Conference 2025, Finalist in the Design Contest held at VLSID 2026.

My student Daksh achieved second rank in India Region in the CSAW 2025 Embedded Security Challenge (ESC).

My students' team (Daksh Pandey and Ketan Anand Roy) achieved second rank at the ISEA Bootcamp and Hackathon on Hardware Security 2026, held at the IIT Kanpur.

My research on transliteration of Modi script to Devanagari was covered by IndiaToday and HinduBusinessLine.

My research on AI-Powered Translation of Text into Fashion Outfits was covered by TimesOfIndia and India Education Diary.

My research on mobile phone detection was praised by Education Minister of India (1, 2, 3). Also, it was covered by HinduBusinessLine, TimesOfIndia, AndhraJyothy, TelanganaToday & TheHansIndia.

My research on 3D SRAM for HPC systems has featured on NextPlatform.

Yash Jain and Vishu Saxena, whom I advised for BTech project, won the best BTech project award in ECE Department, IIT Roorkee in 2023.

Yash and Vishu also won the “Best Innovation” award in the “Professional Development and Innovation Award” scheme at IIT Roorkee (This award was established by 1983 batch alumni of IITR).

My PhD student Maruthi Inukonda received Best Cloud Innovator of the Year 2021 in Cloud Management from Cloud Computing Innovation Council of India.

My MTech student Vipul Agarwal received Prof. Kumkum Garg Gold Medal and Mrs. Chander Mohini Kapoor and Jeewan Kapoor Merit Prize in IITR's Convocation 2025.

Received *Distinguished Contribution* rating at ORNL based on 2013-2014 performance appraisal. This rating recognizes the top 10 percent of staff and is the topmost rating for research staff.

Received *Outstanding Contribution* rating at ORNL based on 2014-2015 performance appraisal.

Google Scholar Citations: 10,600. h-index: 47. i10-index: 114.

ECpE Fellowship of $2500 and Peer Research Award of $200 from ISU

Sumer Chand Jain Scholarship of INR 10,000 at IIT Roorkee.

Silver Medal for getting highest GPA in Electronics batch of year 2008 in ECE department, IIT Roorkee.

Silver Medal for best B.Tech project in Electronics batch of year 2008 in ECE department, IIT Roorkee.

Best Student Award from High School (named MHS, Jaipur) in 2004. Topper throughout school.

Invited talks/panel-discussions/Tutorials

Tutorials: VDAT 2025.

Panelist in: ”Leadership Dialogue on AI in Mobility: Accelerating the Future of Intelligent Transport” session, held under the India AI Impact Summit 2026. ‘HPC and AI: two sides of same coin’ by ACM India and TCS Research 2023.

Conferences: 1. ISC Conference (Germany) 2016. ISC is the biggest HPC conference in Europe with nearly 3000 attendees. 2. PARCOMPTECH (Bengaluru) Organized by C-DAC and MeitY 2017.

Companies: Xilinx (Hyderabad) 2019, Intel (Bangalore) 2017.

Universities: New York University, University of Michigan, IISC Bengaluru, VelTech University (Chennai), MSRIT (Bengaluru), NIT Jalandhar, Anurag Group of Institutions (Hyderabad), DIT University (Dehradun), JIIT (Noida).

Hospitals: AIIMS Rishikesh (In the webinar on ”Artificial Intelligence for Medicine” under the aegis of Telemedicine Society of India, Uttarakhand Chapter).

Reviewer for Proposals submitted to: Technology Development Board (DST).

Funded projects (all as sole-PI)

Research Projects		
Agency	**Title/duration**	**Amount**
SERB Start-up Research Grant	Secure & Reliable Non-volatile Memories for Ultra-low Power Applications (2017-2020)	45 Lakh
SERB Core Research Grant	Designing Secure and Robust Artificial Intelligence (AI) Algorithms and Accelerators (2023-2026)	25 Lakh
Semiconductor Research Corporation (USA)	Designing Efficient Hardware Accelerators for Autonomous Driving Vehicles (2018-2021)	25 Lakh
Scheme for Promotion of Academic and Research Collaboration (SPARC)	Spintronics based Hardware Security Primitives [Awarded to Dr B.K. Kaushik. I am continuing this project after his demise]	57 Lakh
Scheme for Promotion of Academic and Research Collaboration (SPARC)	Artificial Intelligence using spintronics based learning enabled devices (AISLE) [Awarded to Dr B.K. Kaushik. I am continuing this project after his demise]	74 Lakh
Qualcomm Faculty Award	2024-onwards	12.5 Lakh
Qualcomm Innovation Fellowship	2025-26	10 Lakh
Intel PhD Fellowship (for any of my student)	2017-2021	22 Lakh
Startup Grant at IITH and IITR		25 Lakh
Consultancy Projects		
TCS	Architecture for Accelerating Al workloads (2022-2026)	45 Lakh
Bosch	Exploration of Generative AI for Industrial Synthetic Data Generation (2024-2025)	8 Lakh

Patents

Maruthi S. Inukonda, Jessy Bondla, Arjun Reddy, and Sparsh Mittal. Methods and systems for privacy-preserving federated continuous internet forensics, 2023. Indian Patent 202,341,041,491 (filed on 19 June 2023, Granted on 21-July-2025 as patent number 568893).

Ratul Kishore Saha, Manoj Karunakaran Nambiar, Rekha Singhal, and Sparsh Mittal. Method and system for content-aware deployment of artificial intelligence workloads in edge-cloud ecosystem, 2025. Indian Patent 202,521,042,858 (filed on 2 May 2025).

Arghyajoy Mondal, Rajdeep Samantha, Ashwin Krishnan, Rekha Singhal, Manoj Nambiar, and Sparsh Mittal. Method and system of energy estimation for running an application, 2026. Indian Patent 202521096179 (filed on 06-Oct-2025).

Arghyajoy Mondal, Rajdeep Samantha, Ashwin Krishnan, Rekha Singhal, Manoj Nambiar, and Sparsh Mittal. System and method for data shifting in a hardware tiled systolic array accelerator, 2026. Indian Patent 202621028895 (filed on 11 March 2026).

Aaditi Kapre, Shruti Kunde, Sparsh Mittal, and Rekha Singhal. System and methods for hyperspectral image processing in remote sensing using reflexivity based approximate computing, 2023. Indian Patent 202,321,078,005 (filed on 16 November 2023).

Publications

Journal papers are numbered [J1], [J2], etc.; Conferences papers as [C1], [C2], etc. and Review papers as [R1], [R2], etc. Papers with (Corresponding Author)mark are where I am corresponding author.

Onkar Kishor Susladkar, Tushar Prakash, Gayatri Deshmukh, Kiet A Nguyen, Jiaxun Zhang, Adheesh Sunil Juvekar, Tianshu Bao, Lin Chai, Sparsh Mittal, Inderjit S Dhillon, and Ismini Lourentzou. Best of both worlds: Multimodal reasoning and generation via unified discrete flow matching. In *International Conference on Machine Learning (ICML) (Core A*)*, South Korea, 2026. [C1] (Corresponding Author).

Abhishek Goel, Cheena Singhal, Sparsh Mittal, and Sudeb Dasgupta. A 252.8 GOPS and 10.1 TOPS/W Split DP-8T SRAM-Based Analog CIM Macro for 9-Bit Signed MAC Operations with High Signal Margin. *IEEE Transactions on Very Large Scale Integration Systems (VLSI)**(Q2 journal)***, 2026. (Impact Factor 3.1) [J1] Link.

Vishesh Mishra, Mahendra Rathor, Sparsh Mittal, and Urbi Chatterjee. A novel and attack-resilient quad-pixel cmos image sensor-based puf. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, India, 2026. [C2] (Corresponding Author).

Aabid Amin Fida and Sparsh Mittal. Enhancing robustness of convolutional spiking neural networks on memristive hardware. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, India, 2026. [C3] (Corresponding Author).

Daksh Pandey, Ketan Roy, Vishesh Mishra, and Sparsh Mittal. Side-channel based attack optimization and reverse engineering for ml-kem hardware. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2026.

Arghyajoy Mondal, Rajdeep Samanta, Ashwin Krishnan, Sparsh Mittal, Rekha Singhal, and Manoj Nambiar. Boosting energy efficiency of systolic arrays using hardware tiling and johnson counter based data feeding scheme. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2026.

Priyanshu Tyagi, Karunakar Reddy Mudhi Reddy, and Sparsh Mittal. A 90.29-tops/w sram-based digital cim macro featuring novel input sparsity-aware addition and low overhead 30t adder for efficient inference. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, India, 2026.

Abhishek Goel, Cheena Singhal, Ishant Bisht, Sparsh Mittal, and Sudeb Dasgupta. A 785 gops and 3.1 tops/w high-density hybrid rom-sram cim accelerator for few-shot person identification. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI) 2026*, India, 2026.

Rhythm Patel, Priyanshu Tyagi, Sparsh Mittal, and Rekha Singhal. An fpga-based int8 mobilenetv2 accelerator integrated with a risc-v soc for real-time semiconductor wafer defect detection. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI) 2026*, India, 2026.

Prasun Singhal and Sparsh Mittal. Three-timescale reward-modulated plasticity in a two-neuron coupled lif architecture for autonomous obstacle avoidance. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI) 2026*, 2026.

Aabid Amin Fida and Sparsh Mittal. A novel digital implementation of fpga-based reservoir computing using dynamic nodes. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2026.

Rohit Singh Nitwal, Sparsh Mittal, and Manav Aggarwal. Dual-source attention-guided frequency-spatial ensemble attacks for cross-architecture adversarial transferability. *Journal of Systems Architecture* ***(Q1 journal)***, 2026. (Impact Factor 4.1) [J2] Link.

Abhishek Goel, Cheena Singhal, Sparsh Mittal, and Sudeb Dasgupta. A DS-SWL 6T SRAM-Based High-Throughput Hybrid CIM Architecture for 8-Bit MAC Operation in 2-cycle with Time-Domain Weight Categorization. In *IEEE International NEWCAS Conference*, Canada, 2026. [C4] (Corresponding Author).

Vishesh Mishra, Sparsh Mittal, and Urbi Chatterjee. PowerShift: Leveraging Power-Aware Weight Approximations for Neural Network Acceleration. In *The International Conference on VLSI Design (VLSID)*, 2026. [C5] (Corresponding Author).

Shresth Shankhdhar, Ansh Bharadwaj, Vishesh Mishra, and Sparsh Mittal. SilentBite: a Novel LLM-Based Framework for Automated Hardware Trojan Insertion. In *IEEE International Symposium on Circuits and Systems (ISCAS)*, Shanghai, China, 2026. [C6] (Corresponding Author).

Aabid Amin Fida and Sparsh Mittal. An efficient onchip adaptation technique for maml models on memristive hardware. In *IEEE International Symposium on Circuits and Systems (ISCAS)*, Shanghai, China, 2026. [C7] (Corresponding Author).

Priyanshu Tyagi, Dhayan Dhananjaya Senanayake, and Sparsh Mittal. A 38.21-tops/w sram-based digital cim macro for bit-level sparsity-aware mac operations. In *International VLSI Symposium on Technology, Systems and Applications (VLSI TSA)*, Taiwan, 2026. [C8] (Corresponding Author).

Cheena Singhal, Abhishek Goel, Palak, Kartik Shahi, Sparsh Mittal, and Sudeb Dasgupta. A 1781 gops compute-in-memory cim macro with a novel dual-port 8t sram for implementing multiple logical operations. In *International VLSI Symposium on Technology, Systems and Applications (VLSI TSA)*, Taiwan, 2026. [C9] .

Gayatri Deshmukh, Somsubhra De, Chirag Sehgal, Jishu Sen Gupta, and Sparsh Mittal. Dressing the Imagination: A Dataset for AI-Powered Translation of Text into Fashion Outfits and A Novel NeRA Adapter for Enhanced Feature Adaptation. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, USA, 2026. [C10] (Corresponding Author).

Vandan Gorade, Rekha Singhal, Neethi Dasu, Sparsh Mittal, K C Santosh, and Debesh Jha. L2GNet: Optimal Local-to-Global Representation of Anatomical Structures for Generalized Medical Image Segmentation. In *AI for Medicine and Healthcare (AIMedHealth) (Workshop organized with AAAI 2026)*, Singapore, 2026. [C11] .

Onkar Susladkar, Rohit Pawar, Chirag Sehgal, Samaksh Ujjawal, and Sparsh Mittal. Confidence Through Parallel Attention for Depth and Uncertainty Estimation in Dynamic Environments. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, USA, 2026. [C12] (Corresponding Author).

Priyanshu Tyagi, Rhythm Patel, Sparsh Mittal, and Rekha Singhal. An FPGA-Based Secure and Privacy-Aware RISC-V SoC with a CNN Accelerator for Edge AI. In *The International Conference on VLSI Design (VLSID)*, 2026. [C13] (Corresponding Author).

Priyanshu Tyagi, Aayush Gautam, and Sparsh Mittal. A 98.5-TOPS/W and 0.73-TOPS/mm2 Digital Compute-In-Memory Macro with a Novel 6T+MUX Bitcell For Sparsity-Aware MAC Operations. In *The International Conference on VLSI Design (VLSID)*, 2026. [C14] (Corresponding Author).

Harsh Raj Thakur and Sparsh Mittal. A transistor-level implementation of radix-2 and radix-4 fft using logical effort. In *The International Conference on VLSI Design (VLSID)*, 2026. [C15] (Corresponding Author) Link.

Cheena Singhal, Abhishek Goel, Sparsh Mittal, and Sudeb Dasgupta. A 146 GOPS and 7.2 TOPS/W 6T SRAM-Based Analog CIM Macro using Bit-splitting for 8-Bit MAC Operations. In *The International Conference on VLSI Design (VLSID)*, 2026. [C16] (Corresponding Author).

Rohit Singh Nitwal, Manav Aggarwal, and Sparsh Mittal. FUSE: Frequency Spatial Ensemble Attack with Dual Source Attention. In *International Conference on AI-ML Systems (Short-paper)*, 2025. [C17] "Best Poster Award" (Corresponding Author).

Arnav Singh, Soikat Das, and Sparsh Mittal. k-Sparse Audio Adversarial Attack: Budget-Constrained Audio Perturbations with Total Variation Regularization. In *International Conference on AI-ML Systems (Short-paper)*, 2025. [C18] "Best Poster Award" (Corresponding Author).

Onkar Susladkar, Jishu Sen Gupta, Chirag Sehgal, Sparsh Mittal, and Rekha Singhal. MotionAura: Generating High-Quality and Motion Consistent Videos using Discrete Diffusion. In *International Conference on Learning Representations (ICLR)* ***(Core A* rank conference)***, 2025. [C19] (Corresponding Author).

Tushar Prakash, Onkar Susladkar, Sparsh Mittal, and Inderjit Dhillon. Pyramidal Spectrum: Frequency-based Hierarchically Vector Quantized VAE for Videos. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)****(Core A rank conference)***, USA, 2026. [C20] (Corresponding Author).

Sonali Gangwar and Sparsh Mittal. FaEmNet: A Performance-Driven Architecture for Facial Emotion Recognition. In *International Conference on AI-ML Systems*, India, 2025. [C21] (Corresponding Author).

Arghyajoy Mondal, Rajdeep Samanta, Ashwin Krishnan, Sparsh Mittal, Manoj Nambiar, and Rekha Singhal. ArchTune: A predictive Energy Estimation Framework for LLM Inference on Edge Accelerators. In *International Conference on AI-ML Systems*, India, 2025. [C22] .

Abhishek Goel, Cheena Singhal, Dinesh Kushwaha, Sparsh Mittal, and Sudeb Dasgupta. A 6T SRAM Analog CIM Macro for 8-bit MAC with Input/Weight Partitioning for High Signal Margin and Throughput. In *IEEE Asia Pacific Conference on Circuits and Systems*, South Korea, 2025. [C23] .

Vishesh Mishra, Dipesh, Sparsh Mittal, and Urbi Chatterjee. SATGuard: SAT-Driven Countermeasures for Protecting Approximate Circuits from Hardware Trojan. *ACM Transactions on Embedded Computing Systems* ***(Q2 journal)***, 2025. (Impact Factor 2.6) [J3] Link.

Harshal Kausadikar, Tanvi Kale, Onkar Susladkar, and Sparsh Mittal. Historic Scripts to Modern Vision: A Novel Dataset and A VLM Framework for Transliteration of Modi Script to Devanagari. In *19th International Conference on Document Analysis and Recognition (ICDAR)* ***(Core A rank conference)***, China, 2025. [C24] Link (Corresponding Author).

Vishesh Mishra, Sparsh Mittal, and Urbi Chatterjee. Novel Hybrid Probabilistic-Statistical Error Metrics for Approximate Adders. *Journal of Systems Architecture* ***(Q1 journal)***, 2025. (Impact Factor 3.8) [J4] (Corresponding Author) Link.

Priyanshu Tyagi and Sparsh Mittal. A 101 TOPS/W and 3.2 TOPS/mm^2 6T SRAM-Based Digital Compute-in-Memory Macro Featuring a Novel 2T Multiplier. In *Design, Automation and Test in Europe (DATE) (Core B rank conference)*, France, 2025. [C25] (Corresponding Author) Link.

Cheena Singhal and Sparsh Mittal. A 6T SRAM based reconfigurable in-memory XOR/XNOR and accumulation architecture. In *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, Greece, 2025. [C26] (Corresponding Author) Link.

Ratul Kishor Saha, Sparsh Mittal, Rekha Singhal, and Manoj Nambiar. CADAEC: Content-Aware Deployment of AI Workloads in Edge-Cloud Ecosystem. In *International Conference on Performance Engineering (ICPE) (Industry Track) (Core B rank conference)*, 2025. [C27] Link.

Vishu Saxena, Yash Jain, and Sparsh Mittal. A Deep Learning based Approach for Semantic Segmentation of Small Fires from UAV Image. *Remote Sensing Letters*, 2025. (Impact Factor 1.4) [J5] (Corresponding Author) Link.

Aabid Amin Fida and Sparsh Mittal. Stochastic Memristive Devices for Low Cost Learning of Spatiotemporal Signals in Spiking Neural Networks. *IOP Engineering Research Express* ***(Q2 journal)***, 2025. (Impact Factor 1.5) [J6] (Corresponding Author) Link.

Dhayan Dhananjaya Senanayake, Priyanshu Tyagi, Sparsh Mittal, and Rekha Singhal. An SRAM-based Multi-Operand Architecture Implementing Multi-Bit Boolean Functions Using In-Memory Periphery Computing. In *International Conference on VLSI Design*, 2025. [C28] (Corresponding Author) Link.

Onkar Susladkar, Gayatri Deshmukh, Vandan Gorade, and Sparsh Mittal. GRIZAL: Generative Prior-guided Zero-Shot Temporal Action Localization. In *The 2024 Conference on Empirical Methods in Natural Language Processing* ***(Core A* rank conference)***, USA, 2024. [C29] (Corresponding Author) Link.

Vandan Gorade, Sparsh Mittal, Debesh Jha, Rekha Singhal, and Ulas Bagci. Harmonized Spatial and Spectral Learning for Generalized Medical Image Segmentation. In *International Conference on Pattern Recognition (ICPR) (Core B rank conference)*, Kolkata, India, 2024. (Selected for oral presentation, "Best Industry-Related Paper" Award) [C30] (Corresponding Author) Link.

Aabid Amin Fida, Sparsh Mittal, and Farooq Ahmad Khanday. Mott memristor based stochastic neurons for probabilistic computing. *IOP Nanotechnology* ***(Q2 journal)***, 35(29), 2024. (Impact Factor 3.5) [J7] (Corresponding Author) Link.

Onkar Susladkar, Gayatri Deshmukh, Sparsh Mittal, and Parth Shastri. D2Styler: Advancing Arbitrary Style Transfer with Discrete Diffusion Methods. In *International Conference on Pattern Recognition (ICPR) (Core B rank conference)*, Kolkata, India, 2024. [C31] (Corresponding Author) Link.

Anurag Kamal, Vishesh Mishra, Sparsh Mittal, Mahendra Rathor, Chandan Kumar, and Urbi Chatterjee. Sorting Attacks Resilient Authentication Protocol for CMOS Image Sensor Based PUF. In *Asian Hardware Oriented Security and Trust Symposium (AsianHOST)*, Japan, 2024. [C32] Link.

Vandan Gorade, Sparsh Mittal, Debesh Jha, and Ulas Bagchi. SynergyNet: Bridging the Gap between Discrete and Continuous Representations for Precise Medical Image Segmentation. In *Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* ***(Core A rank conference)***, pages 7753–7762, USA, 2024. [C33] (Corresponding Author) Link.

Vandan Gorade, Sparsh Mittal, Debesh Jha, and Ulas Bagchi. Rethinking Intermediate Layers design in Knowledge Distillation for Kidney and Liver Tumor Segmentation. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, Greece, 2024. [C34] (Corresponding Author)**(Oral presentation)**.

Gayatri Deshmukh, Onkar Susladkar, Dhruv Makwana, Sparsh Mittal, and S.C. Teja. Textual Alchemy: CoFormer for Scene Text Understanding. In *Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* ***(Core A rank conference)***, pages 2919–2929, USA, 2024. [C35] (Corresponding Author) One of the 53 papers (out of 847 accepted papers) that was selected for **oral presentation.** Link.

Dhruv Makwana, Gayatri Deshmukh, Onkar Susladkar, Sparsh Mittal, and S.C. Teja. LIVENet: A novel network for real-world low-light image denoising and enhancement. In *Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* ***(Core A rank conference)***, pages 5844–5853, USA, 2024. [C36] (Corresponding Author) Link.

Tushir Sahu, Vidhi Bhatt, Sparsh Mittal, R Sai Chandra Teja, and S Nagesh Kumar. SPEEDNet: Salient Pyramidal Enhancement Encoder-Decoder Network for Colonoscopy Images. In *IEEE International Sym-*

posium on Smart Electronic Systems (IEEE -iSES), New Delhi, India, 2024. [C37] (Corresponding Author) Link.

Divya Aggarwal, R Sai Chandra Teja, and Sparsh Mittal. A stacking ensemble technique to predict speed and distance in 4g and 5g communication datasets. In *IEEE International Symposium on Smart Electronic Systems (IEEE -iSES)*, New Delhi, India, 2024. [C38] (Corresponding Author) Link.

Vibhu, Shivang Bhargav, Vivek Kumar, and Sparsh Mittal. Machine learning based algorithm for shockley-read-hall recombination and augur recombination predictions. In *IEEE International Symposium on Smart Electronic Systems (IEEE-iSES)*, New Delhi, India, 2024. [C39] (Corresponding Author) Link.

Vishesh Mishra, Sparsh Mittal, Nirbhay Mishra, and Rekha Singhal. Security Implications of Approximation: A Study of Trojan Attacks on Approximate Adders and Multipliers. In *Proceedings of International Conference on VLSI Design (VLSID)*, pages 511–516, India, 2024. [C40] (Corresponding Author) **Best student paper award** Link.

Ananya Mantravadi, Siddharth Saini, R Sai Chandra Teja, Sparsh Mittal, Shrimay Shah, R Sri Devi, and Rekha Singhal. CLINet: A Novel Deep Learning Network for ECG Signal Classification. *Journal of Electrocardiology*, 83:41–48, 2024. (Impact Factor 1.3) [J8] (Corresponding Author) Link.

Maruthi S Inukonda, K. Prashanth Thummanapelly, Bheemarjuna Reddy Tamma, and Sparsh Mittal. TEFAR: An Efficient Transparent Finer-grained Encryption of Internet Access Artifacts. In *COMSNETS India Internet Governance Workshop (IIGW)*, 2024. [C41] **Best paper award**.

Dhruv Makwana, R Sai Chandra Teja, and Sparsh Mittal. PCBSegClassNet - A Light-weight Network for Segmentation and Classification of PCB Component. *Expert Systems With Applications* ***(Q1 journal)***, 225:120029, 2023. (Impact Factor 8.665) [J9] (Corresponding Author) Link.

Vandan Gorade, Sparsh Mittal, and Rekha Singhal. PaCL: Patient-aware Contrastive Learning Through Metadata Refinement for Generalized Early Disease Diagnosis. *Computers in Biology and Medicine* ***(Q1 journal)***, 167:107569, 2023. (Impact Factor 7.7), [J10] (Corresponding Author) Link.

Onkar Susladkar, Gayatri Deshmukh, Dhruv Makwana, Sparsh Mittal, S.C. Teja, and Rekha Singhal. GAFNet: A Global Fourier Self Attention Based Novel Network for multi-modal downstream tasks. In *Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* ***(Core A rank conference)***, pages 5231–5240, Hawaii, USA, 2023. [C42] (Corresponding Author) Link.

Onkar Susladkar, Gayatri Deshmukh, S.C. Teja, Sparsh Mittal, and Rekha Singhal. LiBERTy: A Novel Model for Natural Language Understanding. In *Proceedings of ACM International Conference on AI-ML Systems (AIMLSystems)*, pages 1–9, Bengaluru, India, 2023. [C43] (Corresponding Author)**(Best paper award honorable mention)** Link.

Onkar Susladkar, Gayatri Deshmukh, Subhrajit Nag, Ananya Mantravadi, Dhruv Makwana, Sujitha Ravichandran, R Sai Chandra Teja, Gajanan H Chavhan, C Krishna Mohan, and Sparsh Mittal. ClarifyNet: A High-Pass and Low-Pass Filtering Based CNN for Single Image Dehazing. *Journal of Systems Architecture* ***(Q1 journal)***, 132:102736, 2023. (impact factor 5.836) [J11] (Corresponding Author) Link,.

Ananya Mantravadi, Dhruv Makwana, S.C. Teja, Sparsh Mittal, and Rekha Singhal. Dilated Involutional Pyramid Network (DInPNet): A Novel Model for Printed Circuit Board (PCB) Components Classification. In *Proceedings of 24th International Symposium on Quality Electronic Design (ISQED)*, California, USA, 2023. [C44] (Corresponding Author) Link.

Onkar Susladkar, Dhruv Makwana, Gayatri Deshmukh, Sparsh Mittal, S.C. Teja, and Rekha Singhal. TPFNet: A Novel Text In-painting Transformer for Text Removal. In *Proceedings of IEEE International*

Conference on Document Analysis and Recognition (ICDAR) ***(Core A rank conference)***, pages 155–172, California, USA, 2023. [C45] (Corresponding Author) Link.

Mubashir A Kharadi, Sparsh Mittal, and Jhuma Saha. Structural, electronic and optical properties of fluorinated bilayer silicene. *Optical Materials* ***(Q2 journal)***, 136:113418, February 2023. (impact factor 3.754) [J12] Link.

Aabid Amin Fida, Farooq A Khanday, and Sparsh Mittal. An active memristor based rate-coded spiking neural network. *Neurocomputing* ***(Q2 journal)***, 533:61–71, 2023. (Impact Factor 5.779) (Corresponding Author)[J13] Link.

Vishu Saxena, Yash Jain, and Sparsh Mittal. Machine Learning and Polynomial Chaos models for Accurate Prediction of SET Pulse Current. In *Proceedings of IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, pages 1–6, Brazil, 2023. [C46] (Corresponding Author) Link.

Vibhu, Vivek Kumar, and Sparsh Mittal. Machine Learning-based model for Single Event Upset Current Prediction in 14nm FinFETs. In *Proceedings of IEEE International VLSI Design and Embedded Systems conference*, Hyderabad, India, 2023. [C47] (Corresponding Author) Link.

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Initiatives taken and contributions at IITR

My most unique contribution at IITR has been to push the idea of starting a center for offering degree programs on AI and DS. I did the discussion work for this with multiple HODs and director (Prof Chaturvedi). Due to above, I was made the convener of committee formed for initiating artificial intelligence (AI) and Data Science (DS) degree programs. This committee set the ground for creation of the Center for AI/DS (CAIDS) and offering MTech programs in AI/DS.

Later, I was also involved in getting the financial support of Mehta Family for this center and then, this center was renamed to Mehta Family School of DS and AI (MFSDSAI).

I was involved in developing the curriculum of BTech in DS/AI degree offered by MFSDSAI. Right in its first year, this program has attracted top JEE rankers (500). Also, among the DS/AI BTech degrees offered by 8 IITs, our BTech program has attracted the topmost JEE rankers.

I played a major role in developing the proposal for "MTech in VLSI for working professionals" in ECE department, which has been approved by senate in Feb 2021. **This is the first-of-its-kind program in IIT Roorkee.**

Proposed the idea of making department-level profiles on Linkedin for publicity/announcements. This idea was shared with all the departments by the director.

Initiated development of online room-booking system for ECE building

Administrative responsibilities at IITR

TA allocation incharge, where I implemented director's idea of doing TA allocation at department level. This has helped especially new faculty, and this is done probably first time in IIT Roorkee.

Member of ECE DAPC and institute ethics committee. Member of computer center faculty-advisory committee and purchase committee. Convener of faculty-search committee at ECE and MFSDSAI

I was in-charge for converting a room in ECE building to the department server room

Student-guiding and awards/honors to my students

PhD students at IITR: Priyanshu Tyagi, Cheena Singhal, Aabid Amin Fida, Sonali Gangwar, Harsh Raj Thakur, Snigdharani Swain, Akhil Bhogal

MTech students: Sagar, Divyansh, Soikat Das, Karunakar Reddy, Maddi Sai Sushmita, Prasun Singhal, Sadashiv Chinmad, Ankita Rajshri, Avinash Kumar, Aniket Srivastava, Brijesh Yadav

PhD Alumni: Subhrajit Nag, Maruthi S. Inukonda

MTech Alumni: Dinesh Buswala, Rajat Saini, Poonam Rajput, Nandan Jha, Satanu Pattanayak, Srinivas Reddy, Pankaj Berule, Vibhu, Priyansh Singh, Krishna Chaitanya, Prahlad, Dhayan Dhananjaya Senanayake, Vipul Agrawal, Ankit Rathore, Vipin Jayprakash, Ajay Joseph, Aryan Sharma.

Awards to my students: Kumud Lakara (UG intern in our lab) was selected as one among the 9 interns selected as the best by the SPARK committee (IIT Roorkee) in year 2021. Subhrajit selected for IITH-Swinburne joint PhD program.

Courses Taught

Executive Courses: Accelerators for Deep Learning (Cloudx), Accelerators for AI (Coursera), Computer Architecture for AI (short-term course to Qualcomm), RISC-V Assembly Language Programming (short-term course to Qualcomm), etc.

Computer architecture (2016 to 2024) (largest class strength: 164)

Digital Logic (2023 to 2024)

Hardware architectures for deep learning 2019, 2021, 2023 (class strength: 70)

Advanced Computer Architecture (2017-2018) (largest class strength: 55)

Advanced memory system architecture (2016) (class strength: 12)

Shared teaching: Computer vision, Scaling to Big Data, Operating system, Computers and network security

Professional Activity and Outreach

Served as a reviewer for the project proposal submitted to the Technology Development Board.

Gave a talk on "System Engineering for Defense Applications" to 20+ scientists from DRDO (Feb-2026).

Gave a talk in the International Conference on Digital Horizons: Uses and Challenges of AI in Manuscripts and Historical Records Reading, organized by Rajasthan Oriental Research Institute, Bikaner 2026.

Presented cyber-security lessons at IIT Kanpur in the Cyber-commandos internship program.

I was an associate editor of Elsevier's Journal of Systems Architecture for 2+ years.

Organized an online SPARC workshop on "Spintronic Devices and Hardware Security" in Feb 2025. We had 3 speakers from abroad and 4 from different IITs in India.

Organized an online SPARC workshop on "Neuromorphic Computing and Computing in memory" in July 2025. We had 1 speaker from abroad and 2 from different IITs in India.

Organized a SPARC workshop at IITR on "Spintronic Devices" in December 2024, with Prof Prem (NTU) and Prof P.K. Muduli (IIT Delhi) as the experts.

Articles in popular media: Brain drain to Brain gain (Eduvoice), Importance of Coding (BusinessWorld).

My research has featured on Hindu businessline Phys.org, TheMemoryGuy, ScientificComputing, TechEnablement, InsideHPC (1, 2, 3, 4, 5, 6, 7,) Primeur Magazine, StorageSearch (1, 2, 3), HPCWire (1, 2,3, 4, 5), TechDecoded Data-Compression.info, and ReRAM Forum technical websites

Reviewed proposals for three European research-funding agencies (France, Switzerland and Austria).

TPC member of: VLSID 2020, IPDPS 2022, ICPP 2022, NAS 2022, SC 2023, WACV 2024, CCGrid 2024, HiPC 2024, Cluster 2024 and 2025, SC 2024, ISC 2025, IEEE ICEE 2025. Track chair in the iSES 2024 conference and VLSID 2025 and 2026 conferences.

Reviewed MS or PhD thesis of students from IIT Kanpur, IIT Madras and IIIT Hyderabad.

Acted as judge in the startup expo 2022, Smart Indian Hackathon 2022, Defence India Startup Challenges Disc X 2024.

Reviewer for ACM: Computing Surveys, TACO, JETC. IEEE: Trans. on AI, CAL, Trans. on VLSI, Intelligent Systems, ISVLSI, ICCV, Trans. on Image Processing, JETCAS, Trans. on Computers (TC), TCAD, HiPC student research symposium. Elsevier: SUSCOM. Springer: Cluster Computing, J. of Supercomputing

Gave lectures in the workshops held by TEQIP and TLC (teaching learning center) at IIT Hyderabad.

My talks on research-paper writing and presentations have been attended by 1000+ people.

Taught in computer architecture winter school organized by National Supercomputing mission.

Taught in NCERT-IITR Nurturance Programme for National Talent Search (NTS) Awardees 2022.

Guided students under Ishan Vikas Programme initiated by MHRD (Govt of India).